

Don Ngo

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PROFESSIONAL SUMMARY

Research Software Engineer and Data Scientist with a B.S. in Computer Science and Mathematics and experience developing **machine learning workflows**, **scientific software**, **predictive models**, and **high-throughput data systems** for interdisciplinary research. Published research spanning scientific machine learning, crystallographic computing, and data-intensive scientific discovery. Experienced in Python, R, SQL, supervised and unsupervised learning, statistical modeling, scalable data processing, and translating computational methods for domain scientists.

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, JavaScript, C++, Java
- **Machine Learning:** Scikit-learn, TensorFlow, Keras; regression, classification, clustering, predictive modeling, recommendation systems, association-rule mining
- **Data Science:** Pandas, NumPy, SciPy, Tidyverse, statistical analysis, feature engineering, exploratory data analysis, data cleaning, scientific data pipelines
- **Visualization & Applications:** Matplotlib, Tableau, Shiny, React, Django, Flask
- **Databases & Infrastructure:** PostgreSQL, MySQL, SQLite, Docker, Git, SLURM, High-Throughput Computing, REST APIs
- **Scientific Computing:** Parallelization, vectorization, numerical methods, uncertainty propagation, computational geometry, reproducible research

EXPERIENCE

- **Carnegie Science Earth and Planets Laboratory** Washington, D.C.
Research Software Engineer Jun 2025 – Present
 - **Scientific Software Development:** Lead development of **crysttract**, an open-source R package for high-throughput processing and analysis of Crystallographic Information Files, enabling researchers to process **thousands of crystal structures** in reproducible batch workflows.
 - **Scientific Computing & Algorithm Development:** Implemented algorithms for structural analysis, experimental uncertainty propagation, crystallographic geometry, bond distances, and bond-angle calculations; optimized computational workloads using **parallelization and vectorization**.
 - **Open-Source Research Software:** Released **crysttract** through CRAN and led its accompanying peer-reviewed publication in the *Journal of Open Source Software*.
 - **Applied Machine Learning:** Develop machine-learning workflows for scientific datasets, including **predictive modeling, clustering**, structure-property analysis, and mineral classification.
 - **Geospatial Predictive Modeling:** Develop analytical workflows for predicting mineral occurrence from regional mineralogical relationships and locality data using **association-rule mining**, feature engineering, and interactive geospatial visualization.
 - **Mineral Network Explorer (MiNE):** Architect and develop MiNE, an interactive network-analysis platform supporting graph visualization, data preprocessing, community detection, network metrics, and exploratory analysis of large scientific networks.
 - **Research Collaboration:** Collaborate with scientists across mineralogy, crystallography, geoscience, and planetary science to translate domain-specific research questions into computational methods, software systems, and interpretable analytical workflows.
- **Carnegie Science Earth and Planets Laboratory** Washington, D.C.
EPIIC Research Intern, NSF REU Jun 2024 – Aug 2024
 - **Competitive Research Fellowship:** Selected as **1 of 12 interns from 404 applicants** for the NSF-funded Earth and Planetary Science Interdisciplinary Internship Program.
 - **Scientific Data Pipeline:** Developed an automated R pipeline for extracting, cleaning, standardizing, and analyzing crystallographic data, creating analysis-ready datasets for downstream statistical and machine-learning research.
 - **Computational Analysis:** Implemented computational geometry and structural-analysis methods to investigate relationships among crystal structures, chemical composition, and mineral properties.
 - **Independent Research:** Rapidly acquired crystallographic domain knowledge through scientific literature and translated research requirements into reproducible computational workflows and software implementations.

SELECTED MACHINE LEARNING & DATA SCIENCE PROJECTS

- **Mineral Predictive Mapping:** Developing machine-learning and association-analysis workflows for predicting mineral occurrence from regional mineralogical relationships, locality data, and geospatial evidence.
- **Scientific ML for Crystalline Materials:** Developing predictive modeling and clustering workflows to identify structure-property relationships and scientifically meaningful groups within complex crystalline-material datasets.
- **Hood College Hockey Analytics Platform:** Built a full-stack analytics application using **PostgreSQL, Django, and React** to ingest wearable and performance datasets and provide coaches with interactive analysis of player performance, fatigue, and trends.
- **Movie Recommendation System:** Implemented personalized recommendation models using Python, Flask, and React with **content-based filtering** and **collaborative filtering** across a dataset of more than 10,000 movies.

SELECTED PUBLICATIONS

- **Journal of Open Source Software:** Ngo D, Hübner JM, Spintzer M, Morrison SM, Prabhu A (2026). *crysttract: A Crystallography Package in R for .cif Data Processing*. Journal of Open Source Software, 11(122), 9529. doi:10.21105/joss.09529
- **PNAS Nexus:** Li Y, Prabhu A, Hou B, Wong ML, Xing J, Ngo D, Xu B, Hazen RM (2026). *Mineral biosignature identification from Raman spectroscopy using machine learning*. PNAS Nexus, 5(7).
- **Journal of Applied Crystallography:** Prabhu A, Ngo D, Morrison SM, Hübner JM (2026). *Crystrack in Action: Advanced CIF Batch Processing for Structural Analysis in Materials Science and Mineralogy*. Manuscript in review.

EDUCATION

- **Hood College** Frederick, MD
B.S. in Computer Science & Mathematics Aug 2021 – May 2025
Relevant Coursework: Machine Learning, Data Science, Data Mining, Statistics, Advanced Data Structures & Algorithms, Numerical Analysis, Linear Algebra, Database Management Systems, Software Engineering, Operating Systems

HONORS & AWARDS

- Outstanding Student Presentation Award (OSPA) American Geophysical Union
Awarded for an outstanding eLightning presentation in the AGU 2024 Informatics Session. Dec 2024
- Margaret Ann Jones Hayes '62 Award in Mathematics Hood College
Recognized for excellence in the application of computing to mathematics. May 2024

EXTRACURRICULAR ACTIVITIES

- **Returning Presenter, American Geophysical Union Conference** AGU '24, '25
Presented research on scientific software, crystallographic data, and computational methods Dec 2024 – Dec 2025
- **Participant, Planetary Data Science Datathon** Rutgers University
Collaborated with an interdisciplinary team to develop solutions for planetary-science challenges Nov 2025
- **Vice President** ISF Youth Group
Organized community events, managed financial records, and led youth engagement initiatives Feb 2025 – Present
- **Treasurer, Commuter Council** Hood College
Managed organizational finances, budgeting, and financial reporting Aug 2024 – May 2025
- **Volunteer Tutor** Kumon of Urbana, MD
Tutored Pre-K through 12th-grade students in mathematics and reading Mar 2017 – May 2019